

LinkForge — Developer, GitHub & Deployment Runbook

Complete Engineering Guide: Windows Docker Setup, GitHub Workflow, Remote VPS Deployment & Extension Signing (v2.4.2)

1. Fresh Installation on Windows via Docker Desktop & GitHub

Follow this simple 4-step walkthrough to deploy a fresh instance of LinkForge on any Windows PC from your GitHub repository:

Step 1: Prerequisites	Ensure Docker Desktop for Windows (with WSL2 enabled) and Git for Windows are installed and running.
Step 2: Clone Repository	Open PowerShell or Windows Terminal and clone your GitHub repository into your chosen folder: git clone https://github.com/siegewang/Linkforge.git cd Linkforge
Step 3: Launch in Docker	Build and start the container in detached background mode: docker compose up -d --build
Step 4: Open LinkForge	Navigate to http://localhost:5006 (or http://127.0.0.1:5006) in Chrome, Edge, or Firefox. The database, folders, and scheduler initialize automatically!

2. System Architecture & Core Micro-Engines

Component	File / Service	Key Responsibilities
Web App & API	app.py, blueprints/	Flask framework, ProxyFix middleware for SSL reverse proxies, REST APIs for browser extension, PWA share targets.
Neural Link Engine	services/scraper.py	Full-article extraction, offline readability HTML & WebP packaging, auto-tagging, SQLite WAL database storage.
Video & Transcripts	services/videos.py	YouTube transcript ingestion, timestamped navigation, auto-suggest categorization, video bookmarking.
Book Vault Engine	services/books.py	Autonomous Shelfmark grabbing, Hardcover.app API integration, EPUB streaming, keyword filtering & sorting.
LinkForge Pulse	services/pulse.py	AI interest cluster synthesis, multi-source RSS intelligence feeds, real-time news aggregations.
Updater & Backups	services/updater.py services/backup.py	Zero-terminal 1-click GitHub live updates, atomic SQLite snapshots, 12-table CSV exports, full archive packaging.

3. Project Directory Structure

/blueprints/	Modular routes: dashboard.py (home, books, pulse), links.py, videos.py, admin.py, notes.py
/services/	Core background engines: books.py, pulse.py, backup.py, nuke.py, scraper.py, updater.py, db.py
/templates/	Jinja2 UI templates: books.html, pulse.html, admin_nuke.html, admin_extensions.html, base.html
/data/	Persistent volume: linkforge.db (SQLite), /archives/ (offline web cache), /books/ (downloaded EPUBs), /backups/
/dashforge-extension/	Universal Chrome/Edge/Firefox WebExtension source code and manifest.
docker-compose.yml	Container definition, port mapping (5006:5000), volume bindings, healthcheck, and auto-restart policy.

4. How to Update Your GitHub Repository

Whenever you or Antigravity complete code changes on your local PC, push them to GitHub with this standard 3-command workflow:

```
# 1. Navigate to the project folder
cd X:\.config\docker\linkforge
# 2. Stage all modified files and commit with a description
git add .
git commit -m "Describe what you changed here (e.g. Add dark mode toggle)"
# 3. Push to your GitHub repository
git push origin main
```

100% Privacy Guarantee: The .gitignore configuration strictly excludes your /data/ folder, SQLite databases (*.db), downloaded EPUBs, cached article archives, and private secrets. Pushing to GitHub will **NEVER** upload your bookmarks, personal notes, or login credentials.

5. Setting Up Dad's Computer / Remote Devices (Step-by-Step)

Step 1: Open LinkForge	Open any browser on his PC, tablet, or phone and visit https://linkforge.okapitek.uk (or Tailscale IP http://100.64.194.104:5006).
Step 2: Install as PWA App	In Chrome or Edge, click the Install LinkForge icon in the address bar (monitor with down arrow). On iPad/iPhone, tap <i>Share</i> → <i>Add to Home Screen</i> .
Step 3: Install Extension	On his desktop browser, visit /admin/extensions . For Firefox: click 1-Click Direct Install (Firefox XPI) . For Chrome/Edge: unzip dashforge-chrome.zip and Load Unpacked in chrome://extensions.
Step 4: Pin Favorite Sites	On the Home page, click Add Pinned Shortcut to pin his daily sites (BBC News, weather, sports, banking).
Step 5: Book Vault & EPUBs	Show him the Book Vault page. Search any author (e.g. <i>Griff Hosker</i> , <i>Bernard Cornwell</i>), click Auto-Grab Book , and show him 1-click EPUB downloads for his e-reader or tablet.

6. Remote VPS Deployment, Cloudflare Tunnel & Live Updates

Cloudflare Tunnel Configuration (linkforge.okapitek.uk):

In your Cloudflare Zero Trust / Tunnel dashboard, add a Public Hostname rule:

- **Subdomain:** linkforge • **Domain:** okapitek.uk
- **Type:** HTTP • **URL:** localhost:5006 (or linkforge:5000 if sharing Docker bridge)
- **Note:** LinkForge includes ProxyFix middleware, ensuring HTTPS redirects, PWA Web Share Target, and client IPs operate with zero configuration over Cloudflare.

How Future Updates Work (Zero-Terminal / 100% In-Browser):

1. You push new code to GitHub from your PC (git push origin main).
2. Within seconds, your VPS and remote instances detect the new GitHub commit.
3. A glowing badge **UPDATE AVAILABLE** appears in the left sidebar.
4. In Settings, click '**Update LinkForge**' — LinkForge pulls the update and restarts in 2 seconds with zero data loss!

7. Firefox Extension (.XPI) Signing & In-App Distribution

Mozilla Firefox enforces strict cryptographic signatures on all add-ons. To distribute your LinkForge extension to users on standard release Firefox with zero security warnings and 1-click installation, sign the add-on through Mozilla's free self-distribution portal and save it to the LinkForge static downloads directory.

Step 1: Mozilla Portal	Go to https://addons.mozilla.org/developers/addon/submit/distribution and log in.
Step 2: Self-Distribution	Choose ' On your own ' (signs the file for direct distribution without publishing to public store).
Step 3: Upload Zip	Upload X:\.config\docker\linkforge\static\downloads\dashforge-firefox.zip.
Step 4: Download Signed XPI	Mozilla's validator verifies the manifest (0 errors) and signs it in ~1-2 min. Download the signed .xpi.
Step 5: Place in LinkForge	Save the file to replace: X:\.config\docker\linkforge\static\downloads\dashforge-firefox.xpi.
Step 6: 1-Click Install	Now, visiting /admin/extensions allows anyone on Firefox to 1-click install with zero warning screens!

8. Cleaning Test Data (The Nuke Data Tab)

To clean any restored test bookmarks on a fresh container while leaving all AI configurations intact:

- 1. Open Admin Panel ➔ **Nuke Data** (/admin/nuke).
- 2. Click '**Preset: Clean Restored Data (Keep AI Settings)**'.
- 3. Type **NUKE** in the confirmation box and click **Confirm Nuke**.
- 4. All test links, videos, and articles are wiped clean, while the Smart Master Switch, YouTube fetchers, and AI routing remain 100% saved.

9. Troubleshooting & Maintenance Commands Cheat Sheet

Action / Problem	Command to Run in PowerShell / Bash
Fresh Install on Windows (Docker + Git)	git clone https://github.com/siegewang/Linkforge.git && cd Linkforge && docker compose up -d --build
Restart LinkForge Container	docker compose restart
Rebuild from scratch (forced)	docker compose up -d --build --force-recreate
View live application logs	docker logs -f linkforge
Discard local unpushed changes	git reset --hard origin/main
Repackage browser extensions	python package_extensions.py
Create manual full system backup	python -c "from services.backup import create_auto_backup; print(create_auto_backup())"